Is "Active Learning for AMR Model Retraining" Already Done?

Novelty check · Dyen Asif · 24 Sep 2026 · companion to AMR_Plan.pdf

1. Verdict

The core idea is not published. No paper, preprint or indexed abstract found applies active learning to choose which clinical isolates receive lab susceptibility testing so that an AMR prediction model can be retrained or adapted.

But the area around it is crowded. Temporal decay, transfer to new hospitals, semi-supervised learning on unlabelled spectra, and uncertainty estimates have all been published on the same dataset (DRIAMS) in the last two years. A reviewer will ask why active learning, rather than those. The proposal must answer that, and one of its claimed contributions is already taken (§3).

2. What was searched

About 20 queries through a web search engine that indexes PubMed/PMC, bioRxiv, medRxiv, arXiv and publisher sites (ASM, Nature, Oxford, Elsevier, Springer, Frontiers). Terms combined: active learning / uncertainty sampling / label-efficient / semi-supervised / transfer learning / conformal / dataset shift / temporal drift, with antimicrobial resistance / susceptibility / MIC, and MALDI-TOF / DRIAMS / whole-genome / isolates / surveillance.

> **Limit of this check:** it is not a query run inside Scopus or Web of Science, which need institutional access. Before writing "to our knowledge, this has not been done" in a paper, run the query below in Scopus through the FAST library and read every hit's abstract. It takes under an hour. Save the date and hit count — reviewers sometimes ask.

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Scopus:
TITLE-ABS-KEY( ("active learning" OR "uncertainty sampling" OR "query strategy" OR "label-efficient")
  AND ("antimicrobial resistance" OR "antibiotic resistance" OR "antimicrobial susceptibility")
  AND ("MALDI*" OR "mass spectr*" OR "genom*" OR "isolate*") )

PubMed (exclude the education meaning of "active learning"):
("active learning"[tiab] OR "uncertainty sampling"[tiab]) AND ("antimicrobial resistance"[tiab]
  OR "antimicrobial susceptibility"[tiab]) NOT (student*[tiab] OR curriculum[tiab] OR teaching[tiab])
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3. Closest existing work, and what it means for the proposal

Work	What it did	Consequence
Weis et al., *Nature Medicine* 2022 (DRIAMS paper)	Already tested DRIAMS-A over time: train 2015–17, test 2018 → performance dropped; drops grow with temporal distance, worst for *K. pneumoniae*. Partly attributed to instrument maintenance (technical drift).	**Good:** decay exists in your own dataset — premise confirmed. **Bad:** slide 17's contribution 01 ("a measured picture of decay") is already published. Drop it as a contribution; cite it as motivation.
Wiesmann et al., *J Clin Microbiol* 2025	Prospective 18-month decay; says models must be retrained regularly.	Motivation. Unchanged.
Duroux et al., *GigaScience* 2026	Masked autoencoders on DRIAMS for **zero-shot** use at unseen hospitals and years — no new labels at all.	Main competitor for the cross-site question. Your claim must be: a few chosen labels beat zero labels, and by how much.
Genus-level transfer learning, medRxiv 2025; DRIAMS + Taiwan *S. aureus*, *Int J Antimicrob Agents* 2024; MSDeepAMR, *Front Microbiol* 2024	Pre-train elsewhere, fine-tune on local data from a new lab.	These fine-tune on **randomly available** local labels. Active learning decides **which** local labels to get. That is complementary, not competing — use fine-tuning as your base learner.

Work	What it did	Consequence
Semi-supervised MALDI AMR, *Neural Computing and Applications* 2026	Pseudo-labelling and kNN label propagation on DRIAMS: uses unlabelled spectra for free.	Strongest objection: "unlabelled spectra are free — why pay for labels?" Must be a baseline, and ideally combined with active learning.
Corvelo Benz et al. (Borgwardt lab), conformal prediction for MALDI AMR, 2025–26	Statistically valid uncertainty for each prediction.	Ready-made, credible uncertainty for your selection step — and the basis for the improved idea below.
Wu et al., *Nat Commun* 2026; polyacrylamide AL (bioRxiv 2025); whole-cell antibiotic discovery AL (2026)	Active learning to choose which **molecules** to synthesise and test.	Proves the loop works with a wet-lab oracle; different decision. Gap intact.
Yu et al., *PLOS Biology* 2025	Population structure makes more data fail to help.	Supports the diversity arm of selection.
General clinical AI: drift-triggered continual learning across 7 Toronto hospitals; active learning under expert budgets (retinopathy)	Same maintenance problem in other clinical domains.	Cite to show the question is recognised generally but unaddressed for AMR.

4. The improved idea

The change below fixes the two real weaknesses (the lost decay contribution and the "why not semi-supervised" objection) and adds a clinical angle nobody has tested. It is an improvement, not a pivot — the active learning loop stays the centre.

> **Working title:** *Label-efficient maintenance of MALDI-TOF resistance predictors: which isolates should the lab still test?*
>
> **Key observation.** In a real lab using a prediction model, the isolates that get sent for full testing will be the ones the model is unsure about — for patient safety. Those same lab results are the labels available for retraining. So the lab's safety rule *is* an active-learning strategy, whether anyone designed it that way or not. Nobody has asked whether it is a good one.

Revised research questions

1. **RQ1 — Label efficiency over time.** On DRIAMS-A (train 2015–16, quarterly pools 2017, test 2018), how many lab-tested isolates does each selection strategy need to recover 95% of full-retraining AUROC? *(unchanged core)*
2. **RQ2 — Selection under clustered data.** Uncertainty vs spectral diversity vs hybrid, when isolates arrive in clusters of near-identical spectra. *(reworded to match MALDI data)*
3. **RQ3 — Paid labels vs free spectra.** At equal budget, does active learning + semi-supervised learning on the untested spectra beat either alone? *(answers the strongest reviewer objection)*
4. **RQ4 — New hospital.** A model trained on DRIAMS-A is deployed at B, C or D. How few local tests close the gap, compared with zero-shot (Duroux) and random-label fine-tuning? *(was future work)*
5. **RQ5 — Safety routing as selection.** If the lab only tests isolates whose conformal prediction set is uncertain, how good are those labels for retraining compared with the best active-learning strategy — and what does each cost in very major errors along the way? *(new; the novel clinical angle)*

Baselines that must appear (a reviewer will ask for each): random selection · no retraining · full retraining (upper bound) · semi-supervised only (pseudo-labels, no new labels) · fine-tuning on random local labels · zero-shot pretrained model for RQ4.

Revised contributions (replacing slide 17):

1. The first evaluation of active isolate selection for maintaining AMR predictors, under both temporal and cross-site shift, on public data with fixed splits others can reproduce.
2. Evidence on whether paid labels and free unlabelled spectra are substitutes or complements.

3. A test of whether safety-driven lab routing and retraining-driven selection agree — with the clinical cost (very major errors) of each.

Scope warning. Five RQs is a lot for one paper. RQ1–RQ3 make a complete paper for Computers in Biology and Medicine or IEEE JBHI. RQ4 and RQ5 strengthen it toward a higher tier, or become a second paper. Let the supervisors choose.

5. One-sentence novelty claim (for the paper, after the Scopus check)

"Prior work has quantified the decay of MALDI-TOF resistance predictors and proposed zero-shot, transfer and semi-supervised remedies; none has addressed which isolates a laboratory should test to maintain such a model under a fixed budget."

Sources found in this search — Weis et al., Nat Med 2022, doi:10.1038/s41591-021-01619-9 · Wiesmann et al., J Clin Microbiol 2025, doi:10.1128/jcm.01186-25 · Duroux et al., GigaScience 2026, doi:10.1093/gigascience/giaf156 · Genus-level transfer learning of MALDI-TOF MS data, medRxiv 2025.04.25.25326423 · *S. aureus* across DRIAMS and Taiwan, Int J Antimicrob Agents 2024 (PubMed 39244164) · MSDeepAMR, Front Microbiol 2024 · Multi-species multi-antibiotic MALDI-TOF deep learning, Neural Comput Appl 2026, doi:10.1007/s00521-026-12425-0 · Corvelo Benz, Miranda, Chen, Sattler, Borgwardt, conformal prediction for MALDI-TOF AMR, 2026, doi:10.1177/15578666251396558 · Wu et al., Nat Commun 2026 · Cross-molecular active learning for antimicrobial polyacrylamides, bioRxiv 2025.11.07.687243 · Yu, Wheeler, Barquist, PLOS Biol 2025. Read each full text before citing; this table is built from abstracts and summaries.